Worksheet 13: Related Rates

MATH 1271

10/22/2026

1. A water tank has the shape of an inverted circular cone (point down, as in an example from lecture) with a base of radius 6 feet and a depth ("height") of 8 feet. Suppose that water is being pumped into the tank at a constant instantaneous rate of 4 cubic feet per minute. Let r and h be the radius and the depth of the water at a given time, t.

   (a) Draw a picture of the conical tank, including a sketch of the water level at a point in time when the tank is not yet full. Indicate what the variables r and h represent on your figure.

   (b) What equation relates the radius and height of the water? (Hint: Similar triangles!)

   (c) Determine an equation that relates the volume of water, V, in the tank at time t to the depth of the water, h, at that time.

(d) Through differentiation, find an equation that relates the instantaneous rate of change of the water volume, V (with respect to time) to the instantaneous rate of change of the depth of the water.

(e) Find the rate at which the water level is rising at the moment when the water in the tank is 3 feet deep.

(f) When is the water rising most rapidly? When the depth is $h = 3$, $h = 4$, or $h = 5$ feet?

2. A television camera is positioned 4000 feet from the base of a rocket launching pad. The angle of elevation of the camera has to change at the correct rate in order to keep the rocket in sight. In addition, the auto-focus of the camera has to take into account the increasing distance between the camera and the rocket. We assume that the rocket rises vertically.

   (A similar problem is discussed and pictured dynamically https://gvsu.edu/s/9t/. Exploring the applet at the link will be helpful to you in answering the questions that follow.)

   (a) Draw a figure that summarizes the given situation. What parts of the picture are changing? What parts are constant? Introduce variables to represent the quantities that are changing.

   (b) Find an equation that relates the camera's angle of elevation to the height of the rocket.

   (c) Find an equation that relates the instantaneous rate of change of the camera's elevation angle to the instantaneous rate of change of the rocket's height (where all rates of change are with respect to time).

(d) Find an equation that relates the rocket's height to the distance between the camera and the rocket.

(e) Find an equation that relates the instantaneous rates of change of the rocket's height to the instantaneous rate of change of the distance between the camera and the rocket.

(f) Suppose that the rocket's speed is 600 ft/sec at the instant it has risen 3000 feet.

i. How fast is the distance from the television camera to the rocket changing at that moment?

ii. If the camera is following the rocket, how fast is the camera's angle of elevation changing at that same moment?

(g) If from an elevation of 3000 feet onward the rocket continues to rise at 600 feet/sec, will the rate of change of distance with respect to time be greater when the elevation is 4000 feet than it was at 3000 feet, or less? Why?

3. Suppose you are driving along some interstate where the speed limit is 70 miles per hour. A state trooper is hidden about $\frac{3}{100}$'ths of a mile away from the highway, and uses a radar gun to measure the rate of change of a vehicle's distance away from them.

   (a) Draw a picture of this situation and label any quantities you think might be important to this situation.

   (b) Suppose that you are a distance of $\frac{5}{100}$'ths of a mile away from the state trooper, determine the maximum speed at which you can drive so that the state trooper would still measure your speed below the speed limit.